

Smooth slices and superquadratic border determinantal complexity of the permanent

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Abstract

We prove that the border determinantal complexity of the $n \times n$ permanent over $\mathbb{C}$ is $\Omega(n^2 \log n)$. In particular, it is at least $n^2(\log_2 n - 5)/(64e)$ for $n \geq 64$, and at least $3n^2(\log_2 n - 8)/(256eH_2(1/8))$ for $n \geq 512$, where H_2 is binary entropy. The proof combines the matching-polynomial restrictions of OpenAI with Sheshadri's determinantal kernel incidence, using a general linear slice to obtain a smooth homogeneous target. A direct perturbation argument transfers its polar count to coefficient limits of arbitrary affine-linear determinants. We refine the shared matching coefficients by removing their constant block on the critical locus; the remaining coefficient compression also appears in work of Guo. The argument gives corresponding bounds for arbitrary acyclic algebraic branching programs and determines the order of border determinantal complexity of elementary symmetric polynomials. An algebraic transfer retains multiplicities and extends the permanent bounds to algebraically closed fields of characteristic $p > n$. Separate circuit-state bounds allow $O(n^2)$ non-skew products and nonscalar divisions. These restricted circuit consequences do not separate VP from VNP.

1 Introduction and main results

For a polynomial f , its border determinantal complexity $\overline{\text{dc}}(f)$ is the least integer m for which f belongs to the Zariski closure of

$$\mathcal{D}_m = \left\{ \det \left(A_0 + \sum_i x_i A_i \right) : A_i \in \mathbb{C}^{m \times m} \right\}$$

in the space of polynomials of degree at most m . This agrees with coefficientwise approximation over $\mathbb{C}$: $\mathcal{D}_m$ is constructible, and a complex constructible set has the same Euclidean and Zariski closures. Exact determinantal complexity is denoted by dc .

Landsberg, Manivel, and Ressayre [1] proved the quadratic bound $\overline{\text{dc}}(\text{per}_n) \geq n^2/2$. We obtain an unbounded logarithmic improvement in the asymptotic lower bound.

Theorem 1.1. *Over $\mathbb{C}$, for every integer $n \geq 64$,*

$$\overline{\text{dc}}(\text{per}_n) \geq \frac{n^2(\log_2 n - 5)}{64e}. \quad (1)$$

Writing $\gamma = H_2(1/8)$, for every $n \geq 512$ one also has

$$\overline{\text{dc}}(\text{per}_n) \geq \frac{3n^2(\log_2 n - 8)}{256e\gamma}. \quad (2)$$

In particular,

$$\liminf_{n \rightarrow \infty} \frac{\overline{\text{dc}}(\text{per}_n)}{n^2 \log_2 n} \geq \frac{3}{256eH_2(1/8)}.$$

Here $H_2(x) = -x \log_2 x - (1-x) \log_2 (1-x)$. The same bounds hold for exact determinantal complexity.

These explicit inequalities need not exceed $n^2/2$ throughout their stated finite ranges; either may always be combined with the quadratic bound. The first has a shorter parameter calculation, while the second has the larger asymptotic constant.

The proof combines the matching-polynomial restriction and critical-locus bound of [2, Chapter 5, Proposition 5.1 and Lemma 6.1] with the determinantal polar-incidence mechanism of [3, Theorem 1.3]. We include derivations of the needed matching identities and a direct smooth-target transfer proof, so the main argument does not depend on other claims in those works. Guo [5, Proposition 3.1] obtains the same disjointness-matrix compression of the shared matching coefficients. We further use the vanishing of their constant block on the critical locus. Neither the matching selector nor the kernel incidence is claimed as a new construction here. The main result is their composition through a fixed smooth slice, with explicit finite-size estimates.

The elementary-symmetric consequence is in Section 5. Appendix B gives the multiplicity-preserving transfer and its large-characteristic consequence. Appendices C and D concern circuits with a stated budget on non-skew products and nonscalar divisions. The latter statements keep that budget and are distinct from unrestricted circuit lower bounds, such as those in [2, 5].

All critical loci in this note are affine zero sets of all first partial derivatives. A smooth projective hypersurface means a smooth hypersurface scheme, not merely a smooth reduced support. Every degree under consideration is at least two.

2 From critical codimension to a smooth slice

Lemma 2.1. *Let P be a nonzero homogeneous degree- d form on $\mathbb{C}^a$, where $d \geq 2$. Suppose $\text{codim Crit}(P) \geq k \geq 3$. Then a general k -dimensional linear subspace $W \subseteq \mathbb{C}^a$ has the property that $V(P|_W) \subset \mathbb{P}(W)$ is a smooth degree- d hypersurface.*

Proof. If $k = a$, the critical locus is $\{0\}$ and there is nothing to prove. Suppose $k < a$. The Grassmannian $\text{Gr}(k, a)$ has dimension $k(a-k)$. For a smooth point $[x] \in V(P) \subset \mathbb{P}^{a-1}$, the restriction can be singular at $[x]$ only if

$$[x] \in \mathbb{P}(W) \subseteq T_{[x]}V(P).$$

For this fixed point, the choices of W form a Grassmannian of dimension $(k-1)(a-k-1)$. Since the smooth locus of the hypersurface has dimension $a-2$, the incidence of these pairs has dimension

$$(a-2) + (k-1)(a-k-1) = k(a-k) - 1.$$

The projective singular locus has dimension at most $a-k-1$, by homogeneity and Euler's identity. For a fixed singular point, the subspaces containing it have dimension $(k-1)(a-k)$. Thus the other bad incidence also has dimension at most $k(a-k) - 1$. Their images cannot fill the Grassmannian. A general W avoids both and has nonzero restriction, proving the claim. $\square$

Remark 2.2. This argument is necessary: restricting the original gradient and projecting its output is not the same operation as taking the gradient of a restriction. The tangent incidence excludes singularities created by the latter operation. The subspace W will be fixed before any border limit is taken.

3 A border bound for smooth homogeneous targets

We give the precise smooth-target consequence of the kernel-incidence method [3] needed here. The proof uses persistence of simple affine polar points, avoiding a general assertion about specialization of conormal cycles.

Proposition 3.1. *Let f be a homogeneous degree- d form in $k \geq 3$ variables, with $V(f) \subset \mathbb{P}^{k-1}$ smooth. If $\overline{\text{dc}}(f) \leq m$, then*

$$d(d-1)^{k-2} \leq B(m, k), \quad (3)$$

$$B(m, k) = \sum_{a=1}^{k-1} \binom{m}{a} \binom{m-1}{k-1-a} \binom{k-2}{a-1}. \quad (4)$$

Consequently,

$$\overline{\text{dc}}(f) \geq \frac{(k-1)(d-1)}{4e}. \quad (5)$$

Proof. Simple points on the target. Euler's identity and smoothness imply that the first partials of f have no common projective zero. They define a morphism on $\mathbb{P}^{k-1}$ whose pullback of $\mathcal{O}(1)$ is $\mathcal{O}(d-1)$. This morphism is finite: a positive-dimensional fiber would contain a curve on which an ample line bundle had degree zero. Its restriction to $V(f)$ is finite as well.

Choose $k-2$ general constant-coefficient directional derivatives $D_1, \dots, D_{k-2}$. In characteristic zero a general linear section of the image avoids the branch locus and pulls back transversely. The intersection on $V(f)$ therefore consists of

$$\delta = d(d-1)^{k-2}$$

distinct points. Equivalently, this follows by successive general members of the basepoint-free polar system and Bezout. Choose a linear form ℓ nonzero at all these points and rescale each representative to satisfy $\ell = 1$. The resulting points are simple zeros of the square affine system

$$f = 0, \quad \ell - 1 = 0, \quad D_j f = 0 \quad (1 \leq j \leq k-2). \quad (6)$$

Persistence under approximation. Work in the coefficient space of polynomials of degree at most m . The implicit-function theorem at each of the finitely many simple zeros in (6) gives a neighborhood of f in which every polynomial g has a distinct simple zero near each one for the system with f replaced by g . Since $f \in \overline{\mathcal{D}_m}$, choose $g = \det A$ in this neighborhood, where A is an affine-linear $m \times m$ matrix. At the retained zeros, $\nabla g \neq 0$, so $\text{rank } A = m-1$. No convergence of the coefficients of A is required.

A square kernel-incidence system. Write $\hat{A}(z, x) = zA_0 + \sum_i x_i A_i$. For each retained point, choose its unique projective left and right kernel lines, represented by u and v . Choose a linear map $\Lambda : \mathbb{C}^m \rightarrow \mathbb{C}^{m-1}$ injective on the image of A at every one of these finitely many points. Each is a nonempty open condition on Λ , so a common choice exists. On

$$\mathbb{P}_{[z:x]}^k \times \mathbb{P}_{[u]}^{m-1} \times \mathbb{P}_{[v]}^{m-1}$$

consider the following system:

$$u^\top \hat{A} = 0, \quad \Lambda \hat{A} v = 0, \quad u^\top (D_j A) v = 0, \quad \ell(x) - z = 0. \quad (7)$$

There are respectively m , $m-1$, $k-2$, and one equations. Their multidegrees are $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$, and $(1, 0, 0)$. The number of equations equals the ambient dimension.

Isolation is local, not a genericity assumption on A . Fix a retained point and work in the affine chart $z = 1$. After permuting rows and columns, write

$$A = \begin{pmatrix} B & c \\ r & s \end{pmatrix}, \quad \det B \neq 0.$$

In suitable kernel charts the full left equations solve $u^\top = (-rB^{-1}, 1)$ and impose $s - rB^{-1}c = 0$. On this hypersurface, injectivity of Λ on the matrix image persists locally. The reduced right equations thus solve $v = (-B^{-1}c, 1)$. In particular, the kernel incidence is locally a graph over $g = 0$. On that graph,

$$\operatorname{adj}(A) = (\det B)vu^\top, \quad D_j g = (\det B)u^\top(D_j A)v.$$

The factor $\det B$ is a unit. Thus the retained simple zeros of the affine polar system lift to reduced isolated solutions of (7). Other components, including components at infinity or of higher corank, do not affect this assertion.

Counting. Multihomogeneous Bezout bounds the number of isolated solutions by

$$[H^k U^{m-1} V^{m-1}] H(H+U)^m (H+V)^{m-1} (U+V)^{k-2}. \quad (8)$$

This remains a bound when there are excess components: perturb the sections of the globally generated line bundles to a proper intersection; each reduced isolated solution persists. Expanding the coefficient in (8) gives (4), and proves (3).

Finally, Vandermonde's identity gives

$$B(m, k) \leq 2^{k-2} \binom{2m-1}{k-1} \leq 2^{k-2} \left(\frac{e(2m-1)}{k-1} \right)^{k-1}.$$

If the binomial coefficient vanishes, (3) is impossible; otherwise the displayed estimates imply

$$m \geq \frac{1}{2} + \frac{k-1}{4e} (2d(d-1)^{k-2})^{1/(k-1)} \geq \frac{(k-1)(d-1)}{4e}.$$

□

Corollary 3.2. *If P is homogeneous of degree $d \geq 2$ and $\operatorname{codim} \operatorname{Crit}(P) \geq k \geq 3$, then*

$$\overline{\operatorname{dc}}(P) \geq \frac{(k-1)(d-1)}{4e}.$$

The same lower bound holds for any polynomial having a nonzero scalar multiple of P as an affine restriction.

Proof. Apply Lemma 2.1 and Proposition 3.1. Affine substitution preserves size- m determinantal representations and commutes with coefficientwise limits. A nonzero scalar is absorbed by scaling one row of the representing matrix. □

4 The permanent specialization and its parameters

We state the matching inputs and their coefficient refinement. Appendix A supplies all the required identities. For a $t \times s$ matrix of variables X , let

$$M_{t,s,d}(X) = \sum_{\substack{I \subseteq [t], J \subseteq [s] \\ |I|=|J|=d}} \operatorname{per}(X_{I,J}).$$

Proposition 5.1 of [2, Chapter 5] states, for $3 \leq d \leq \min(t, s)$,

$$\dim \text{Crit}(M_{t,s,d}) \leq 2^t - 1 + s(d - 2). \quad (9)$$

The refined estimate proved in Appendix A.2 is

$$\dim \text{Crit}(M_{t,s,d}) \leq R_{\text{crit}}(t, d) + s(d - 2), \quad R_{\text{crit}}(t, d) = \sum_{j=1}^{d-1} \binom{t}{\min(j, d-j)} - t. \quad (10)$$

For b disjoint blocks of variables $X^{(h)}$ and nonzero scalars λ_h , put

$$P = \sum_{h=1}^b \lambda_h M_{t,s,d}(X^{(h)}).$$

Its critical locus is the product of the individual critical loci. Consequently,

$$\text{codim } \text{Crit}(P) \geq b(s(t - d + 2) - 2^t + 1). \quad (11)$$

Lemma 6.1 of the same chapter supplies suitable nonzero λ_h such that P is a nonzero scalar multiple of an affine restriction of per_{bt+s-d} . More explicitly, if $r = bt$, partition the r rows into blocks $R_1, \dots, R_b$ of size t . For a primitive d th root ζ , take the constant columns of U to be $t - d$ copies of $\mathbf{1}_{R_h}$ for each h , and

$$\mathbf{1}_{R_1 \cup \dots \cup R_{h-1}} + 2\zeta^j \mathbf{1}_{R_h} \quad (2 \leq h \leq b, \ 0 \leq j < d).$$

Then U has $r - d$ columns, and the required affine matrix is

$$\begin{pmatrix} X & U \\ \mathbf{1} & 0 \end{pmatrix},$$

where the lower block has $s - d$ rows. The common nonzero scalar is $(s - d)!(t - d)!(t!)^{b-1}$. With $\theta = (-1)^{d+1}2^d$, the weights may be written

$$\lambda_1 = \theta^{b-1}, \quad \lambda_h = \theta^{b-h}(1 + \theta)^{h-2} \quad (2 \leq h \leq b).$$

In particular no weight vanishes.

Proposition 4.1 (Parameter bound). *Let $b \geq 1$ and $3 \leq d \leq \min(t, s)$ be integers, with $n_0 = bt + s - d \leq n$. Set*

$$K = b(s(t - d + 2) - R_{\text{crit}}(t, d)).$$

If $K \geq 3$, then

$$\overline{\text{dc}}(\text{per}_n) \geq \frac{(K - 1)(d - 1)}{4e}.$$

One may instead use $K = b(s(t - d + 2) - 2^t + 1)$.

Proof. The product description of the critical locus and (10) give codimension at least K . Use the selector above, Corollary 3.2, and, when $n_0 < n$, pad the permanent specialization with an identity block. $\square$

4.1 Explicit parameter choices

Proof of Theorem 1.1. We first prove (1) using the uncompressed estimate. For $n \geq 64$, set

$$t = \lfloor \log_2 n \rfloor, \quad d = \lfloor t/2 \rfloor, \quad b = \left\lfloor \frac{n}{2t} \right\rfloor, \quad r = bt, \quad s = n - r + d.$$

Then $t \geq 6$, $b \geq 1$, $3 \leq d \leq \min(t, s)$, and $r + s - d = n$. Since $n/2 - t < r \leq n/2$,

$$rs = r(n - r + d) \geq \frac{n^2}{4} - t^2.$$

For $K = b(s(t - d + 2) - 2^t + 1)$, using $2^t \leq n$ gives

$$\begin{aligned} K &= rs \frac{t - d + 2}{t} - \frac{r}{t}(2^t - 1) \\ &\geq \left(\frac{n^2}{4} - t^2 \right) \left(\frac{1}{2} + \frac{2}{t} \right) - \frac{n^2}{2t} = \frac{n^2}{8} - \frac{t^2}{2} - 2t. \end{aligned}$$

This lower bound is greater than three: at $t = 6$ it is at least $512 - 18 - 12$, and $2^{2t}/8 - t^2/2 - 2t$ increases thereafter. Also $K \leq rs$. As $d - 1 \geq (t - 3)/2$, we have

$$(K - 1)(d - 1) \geq \frac{n^2(t - 3)}{16} - \frac{(t^2/2 + 2t + 1)(t - 3)}{2}.$$

For $t \geq 6$,

$$8(t^2/2 + 2t + 1)(t - 3) \leq 8t^3 \leq 2^{2t} \leq n^2.$$

For the middle inequality, check $t = 6$ and note that $((t + 1)/t)^3 < 4$. Consequently

$$(K - 1)(d - 1) \geq \frac{n^2(t - 4)}{16} > \frac{n^2(\log_2 n - 5)}{16}.$$

Proposition 4.1 proves (1).

For (2), put

$$\gamma = H_2(1/8), \quad c = \frac{8^8}{7^7} = 2^{8\gamma}.$$

Let a be the largest integer with $c^a \leq n$, and set

$$t = 8a, \quad d = 2a, \quad b = \left\lfloor \frac{n}{16a} \right\rfloor, \quad r = bt, \quad s = n - r + d.$$

Since $20 < c < 21$ and $n \geq 512$, we have $a \geq 2$ and $b \geq 1$; all hypotheses of Proposition 4.1 hold. The profile in Appendix A.2 gives

$$R_{\text{crit}}(8a, 2a) = 2 \sum_{j=1}^{a-1} \binom{8a}{j} + \binom{8a}{a} - 8a \leq \frac{4}{3} \binom{8a}{a} \leq \frac{4}{3} c^a \leq \frac{4}{3} n. \quad (12)$$

Indeed, the ratios of consecutive terms below $\binom{8a}{a}$ are at most $a/(7a + 1) < 1/7$. For the next inequality, the single a th term in the binomial expansion of $(1/8 + 7/8)^{8a}$ is at most one, so $\binom{8a}{a} \leq (8^8/7^7)^a$.

Write $K = b(s(t - d + 2) - R_{\text{crit}}(t, d))$. As before, $rs \geq n^2/4 - 64a^2$; therefore

$$\begin{aligned} K &\geq \left(\frac{n^2}{4} - 64a^2\right) \left(\frac{3}{4} + \frac{1}{4a}\right) - \frac{n^2}{12a} \\ &= \frac{3n^2}{16} - \frac{n^2}{48a} - 48a^2 - 16a. \end{aligned}$$

For $a \geq 2$, the polynomial error satisfies

$$(48a^2 + 16a + 1)(2a - 1) \leq 96a^3, \quad 48 \cdot 96a^3 \leq 20^{2a} < c^{2a} \leq n^2.$$

The second inequality holds at $a = 2$ and is preserved on increasing a , because $((a + 1)/a)^3 < 400$. It follows that

$$\begin{aligned} (K - 1)(2a - 1) &\geq n^2 \left(\frac{3a}{8} - \frac{11}{48} + \frac{1}{48a}\right) - (48a^2 + 16a + 1)(2a - 1) \\ &\geq n^2 \left(\frac{3a}{8} - \frac{1}{4}\right). \end{aligned}$$

This estimate in particular ensures $K \geq 3$; for example its right side divided by $2a - 1$ is greater than $n^2/8$. Also $K \leq rs$, directly from its definition. Since $a > \log_2 n / (8\gamma) - 1$ and $\gamma < 3/5$,

$$(K - 1)(d - 1) \geq n^2 \left(\frac{3 \log_2 n}{64\gamma} - \frac{5}{8}\right) \geq \frac{3n^2(\log_2 n - 8)}{64\gamma}.$$

Another application of Proposition 4.1 proves (2). Finally $\text{dc} \geq \overline{\text{dc}}$. $\square$

Corollary 4.2 (Unlayered algebraic branching programs). *Let an algebraic branching program mean an arbitrary finite directed acyclic graph with a designated source and sink and affine-linear edge labels over $\mathbb{C}$. Its output is the sum of the weights of all source-to-sink paths. If per_n is a coefficientwise limit of outputs of such programs with at most v vertices, then*

$$v + 1 \geq \frac{n^2(\log_2 n - 5)}{64e} \quad (n \geq 64),$$

and

$$v + 1 \geq \frac{3n^2(\log_2 n - 8)}{256eH_2(1/8)} \quad (n \geq 512).$$

The same bound holds for exact computation. No layering, homogeneity of intermediate computations, or variable-order restriction is assumed.

Proof. In a topological ordering, let A be the weighted adjacency matrix of the program; parallel-edge labels can be added. It is strictly upper triangular, and its path polynomial is

$$e_s^\top (I - A)^{-1} e_t = \det \begin{pmatrix} I - A & e_t \\ -e_s^\top & 0 \end{pmatrix}.$$

This is an affine-linear determinant of size at most $v + 1$. Pad with identity blocks if necessary and take coefficientwise limits. Theorem 1.1 applies. $\square$

The model distinction matters at polynomial scales. For example, Chatterjee, Kumar, She, and Volk [7, Theorems 1.2 and 1.3] obtain different bounds for layered and unlayered branching programs. Corollary 4.2 is stated directly for arbitrary acyclic graphs. It is an immediate consequence of the determinantal statement; it is not a simulation of arbitrary arithmetic circuits by these graphs.

5 An elementary-symmetric control family

The same bridge has a consequence independent of the matching specialization. It provides a useful check on both the degree dependence and the distinction between circuits and determinantal representations. Write $e_{N,d}$ for the elementary symmetric polynomial of degree d in N variables.

Corollary 5.1. *Uniformly for $2 \leq d \leq N$, over $\mathbb{C}$,*

$$\overline{\text{dc}}(e_{N,d}) = \Theta(d(N-d+1)), \quad \text{dc}(e_{N,d}) = \Theta(d(N-d+1)).$$

For $d \leq N-1$, a lower bound is $(N-d+1)(d-1)/(4e)$. An exact upper bound is $(d+1)(N-d+1)+1$.

Proof. The critical locus of $e_{N,d}$ consists of vectors with at most $d-2$ nonzero coordinates; this is a known characteristic-zero fact, recalled in [6]. Here is a direct proof. The identities

$$\sum_i \partial_i e_{N,d} = (N-d+1)e_{N,d-1}, \quad \partial_i e_{N,d} = e_{N,d-1} - x_i \partial_i e_{N,d-1}$$

imply that, at a critical point, all derivatives of e_{d-1} on its nonzero support vanish. If the support has size $p \geq d-1$, repeat the argument on those p coordinates. Each coefficient $p-j+1$ that must be cancelled is positive, hence nonzero in $\mathbb{C}$. Eventually all first derivatives of $e_{p,1}$ would vanish, an impossibility. The converse follows from the degree of each cofactor. Thus the critical codimension is $N-d+2$, and Corollary 3.2 proves the lower bound for $d \leq N-1$. For $d = N$, degree gives $\overline{\text{dc}}(e_{N,N}) \geq N$ and the diagonal matrix gives equality.

For the upper bound, use vertices (j, k) with $0 \leq j \leq d$ and $0 \leq k \leq N-d$. The edge from (j, k) to $(j+1, k)$ has weight x_{j+k+1} , and the edge to $(j, k+1)$ has weight 1, whenever the indicated target exists. Source-to-sink paths correspond bijectively to d -subsets of $[N]$. Their weight sum is therefore $e_{N,d}$. Let A be the weighted adjacency matrix in topological order, with rows indexing sources and columns indexing targets. It is strictly upper triangular. If s, t denote the source and sink, the affine-linear matrix

$$\begin{pmatrix} I - A & e_t \\ -e_s^\top & 0 \end{pmatrix}$$

has determinant $e_s^\top (I - A)^{-1} e_t = e_{N,d}$, by the Schur complement and $\det(I - A) = 1$. Its size is $(d+1)(N-d+1)+1$. These estimates give absolute constants in both Theta statements, including $d = N$. $\square$

6 Scope and limitations

The proof is symbolic; finite computation is not a substitute for any of its asymptotic steps. An accompanying exact checker verifies selector minors, complete specialized permanent identities, small instances of the partition identity behind (9), multihomogeneous coefficient expansions, and the integer parameter estimates. It also verifies the Schur-complement kernel identities. For controls, a smooth four-variable quadric represented by $\det_2$ has polar count $2 = B(2, 4)$; a smooth four-variable cubic section of $\det_3$ has count $12 \leq B(3, 4) = 16$.

The full $n \times n$ determinant has critical codimension four, so Corollary 3.2 applied directly to it gives only a linear bound in n , consistent with its size- n representation. In particular, the smoothness hypothesis must not be omitted.

The result concerns determinantal representations, a model closely related to algebraic branching programs. It supplies no lower bound of the same order for unrestricted arithmetic circuits. For

example, $\sum_{i=1}^N x_i^N$ has a circuit of size $O(N \log N)$, yet Proposition 3.1 gives border determinantal complexity $\Omega(N^2)$. This directly precludes charging the same measure linearly to arbitrary circuit gates.

The obstruction persists for syntactically multilinear circuits, even without repeated squaring. For N a power of two and $d = N/2$, the elementary-symmetric argument supplies a smooth-slice product $(N - d + 1)(d - 1) = \Theta(N^2)$. Yet $e_{N,d}$ has a division-free syntactically multilinear circuit of size $O(N \log^2 N)$. Recursively multiply the polynomials $1 + x_i z$ by fast Fourier convolution on disjoint variable blocks. A block of b variables costs $O(b \log b)$ gates; summing over the balanced recursion proves the bound. Every nonconstant multiplication has disjoint input supports. This standard construction is recalled in [10, Section 5]. It rules out even a charge weakened by any fixed polylogarithmic factor. Its $O(N \log N)$ non-skew products lie outside the $O(N)$ budget of Appendix C; its Fourier gates mix degrees, so this example alone makes no assertion about homogeneous internal gates.

Paired variables remove that remaining distinction for the present *affine* slice measure. Define

$$H_{N,d}(x, y) = [z^d] \prod_{i=1}^N (y_i + x_i z).$$

The same FFT construction is now homogeneous and syntactically set-multilinear in the groups $\{x_i, y_i\}$. In a recursion block I , every coefficient and every Fourier value has degree $|I|$ and uses exactly the groups in I . Additions preserve that group set, while products combine disjoint child blocks. Zero padding may be removed by constant propagation. The size remains $O(N \log^2 N)$. Setting all $y_i = 1$ gives $e_{N,d}$, so at $d = N/2$ the same quadratic smooth-slice product is obtained by an affine restriction. Thus homogeneity and syntactic set-multilinearity together still do not repair a linear or polylogarithmically weakened charge of this affine-slice measure to circuit size. A measure using only homogeneous linear restrictions would be different and would not automatically admit the degree-lowering permanent specialization used above.

There is also an intrinsic limit to this particular slice method. An affine restriction of per_n has degree at most n . A smooth homogeneous restriction of degree at least two can have at most n^2 variables: an extra kernel direction of the affine map would give a projective critical point. Thus the product of degree and smooth-slice dimension is at most cubic in n . The method cannot by itself prove superpolynomial determinantal complexity, let alone $\text{VP} \neq \text{VNP}$. Sheshadri [4] studies the broader degree-times-variables scale of conormal intersection bounds. Here the number of variables of the smooth target is $k = \Theta(n^2)$, while its degree is $d = \Theta(\log n)$. Thus the present $\Theta(kd)$ scale is compatible with that discussion; the quadratic scale for forms whose degree equals their number of variables is a different specialization.

The supplementary verification code checks finite instances of the identities and parameter inequalities. The uniform statements follow from the proofs above and in the appendices.

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A Matching identities and coefficient compression

The finite-module argument and selector reproduce the ingredients of [2, Chapter 5] used above. Between them we give the coefficient refinement, including its relation to [5].

A.1 The critical-locus dimension bound

Put $q = d - 1 < t$. For each nonempty $B \subseteq [t]$, introduce a shared parameter p_B . For a column vector $u \in \mathbb{C}^t$, define

$$Q_i(u; p) = \sum_{\substack{I \subseteq [t] \setminus \{i\} \\ |I|=q}} \sum_{\pi \vdash I} \mu(\pi) \prod_{B \in \pi} (p_B - u_B), \quad u_B = \prod_{j \in B} u_j, \quad (13)$$

where $\mu(\pi) = \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!$. At a matrix X , specialize $p_B = \sum_{a=1}^s \prod_{j \in B} x_{ja}$. Inclusion-exclusion over equalities among column choices gives

$$Q_i(X_{\bullet a}; p(X)) = \partial_{ia} M_{t,s,d}(X).$$

Indeed, $p_B - u_B$ sums the products belonging to columns other than a , and the partition-lattice Möbius coefficient enforces that the q selected rows use distinct such columns.

When the p_B are treated as coefficients, the highest u -degree part of Q_i is

$$\alpha e_q(u_1, \dots, \widehat{u_i}, \dots, u_t), \quad \alpha = (-1)^q q!. \quad (14)$$

To see the coefficient, choose $-u_B$ in every factor of (13). For each fixed I , the sum of $\prod_{B \in \pi} (|B| - 1)!$ over partitions is $q!$: it counts permutations by their cycle partitions. Every other term has u -degree less than q .

Take s column variables $u^{(a)}$ and, for each column, base parameters $c_{j,a}$ for $1 \leq j < q$. Work over the polynomial ring

$$A = \mathbb{C}[p_B, c_{j,a} : \emptyset \neq B \subseteq [t], 1 \leq j < q, 1 \leq a \leq s].$$

Form the quotient of $A[u^{(1)}, \dots, u^{(s)}]$ by $Q_i(u^{(a)}; p) = 0$ and $e_j(u^{(a)}) = c_{j,a}$ for $j < q$. We claim that this quotient is a finite A -module.

Fix a column. Summing (14) over i and using $\sum_i e_q(u_i) = (t - q)e_q(u)$ expresses $e_q(u)$ as a polynomial of u -degree below q , with coefficients in A . Here $\alpha(t - q) \neq 0$. The identity

$$e_q(u_i) = \sum_{j=0}^q (-u_i)^j e_{q-j}(u)$$

then turns each equation $Q_i = 0$ into a monic relation $u_i^q =$ terms of total u -degree below q ; the intermediate e_{q-j} have been replaced by their base parameters. Reducing these powers strictly lowers total column degree. Doing this for every column proves that the finitely many monomials with all exponents less than q generate the quotient over A .

Its Krull dimension is consequently at most $\dim A = 2^t - 1 + s(q - 1)$. The coordinate ring of the critical locus is a quotient: substitute $u^{(a)} = X_{\bullet a}$, $p_B = p_B(X)$, and $c_{j,a} = e_j(X_{\bullet a})$. All defining relations hold, and the map is surjective because it includes every x_{ia} . Dimension cannot increase under this quotient. This proves (9). In particular, the shared parameters need not be independent at a critical point; enlarging to independent base parameters only enlarges the ambient incidence.

A.2 The coefficient image and its constant block

We refine the preceding argument by counting the coefficients of Q_i , rather than all the moments. The same disjointness-matrix coefficient count is used in [5, Section 3]. The additional identity (16)

removes its constant block on the critical locus. We give the details over an algebraically closed field of characteristic zero or characteristic $p > \max(t, s)$.

For $\emptyset \neq S \subseteq [t]$ with $|S| \leq d-1$, put

$$h_S = \sum_{\pi \vdash S} \mu(\pi) \prod_{B \in \pi} p_B, \quad h_\emptyset = 1.$$

For $\emptyset \neq T \subseteq [t]$, $|T| \leq d$, define

$$c_T = \sum_{\substack{S \subseteq [t] \setminus T \\ |S|=d-|T|}} h_S.$$

Collecting coefficients in (13) gives

$$Q_i(u; p) = \sum_{\substack{J \subseteq [t] \setminus \{i\} \\ |J| \leq d-1}} (-1)^{|J|} |J|! c_{J \cup \{i\}} u_J. \quad (15)$$

To see this, choose the blocks supplying a factor $-u_B$ and let J be their union. The partitions of the remaining rows contribute h_S . On J , summing the products of cyclic-order counts over its partitions gives $|J|!$, with sign $(-1)^{|J|}$. Summing over the remaining S proves the identity. The leading coefficients, for $|J| = d-1$, have $c_{J \cup \{i\}} = 1$.

For $|T| = j$, the vector (c_T) is a linear image of $(h_S)_{|S|=d-j}$ under the disjointness matrix on j -subsets and $(d-j)$ -subsets of $[t]$. Its rank is

$$\min \left\{ \binom{t}{j}, \binom{t}{d-j} \right\} = \binom{t}{\min(j, d-j)}.$$

Here is a proof of the rank fact, including its characteristic range. Transpose and complement subsets to reduce to the following assertion: if $a \leq v \leq t-a$ and

$$\sum_{\substack{A \subseteq V \\ |A|=a}} z_A = 0 \quad \text{for every } v\text{-subset } V \subseteq [t],$$

then every $z_A = 0$. Induct on a , the case $a = 0$ being immediate. For distinct i, j , subtract the equations for $U \cup \{i\}$ and $U \cup \{j\}$, where U ranges over the $(v-1)$ -subsets avoiding i, j . On the remaining $t-2$ elements the induction hypothesis applies to the coefficients $z_{C \cup \{i\}} - z_{C \cup \{j\}}$: $a-1 \leq v-1 \leq (t-2) - (a-1)$. They all vanish. The exchange graph of a -subsets is connected, so every z_A has one common value z . The original equation now reads $\binom{v}{a} z = 0$; this binomial coefficient is nonzero in the stated characteristics. This proves the rank formula, a form of the classical inclusion-matrix rank theorem also used in [5].

The change from the independent moments p_S to the h_S is triangular by $|S|$, with nonzero diagonal coefficient $(-1)^{|S|-1}(|S|-1)!$. Thus the h_S are independent coordinates. Different j use different degrees $d-j$, and their images occupy different monomials in (15). Consequently the formal coefficient image has dimension exactly

$$R(t, d) = \sum_{j=1}^{d-1} \binom{t}{\min(j, d-j)}.$$

The $j = 1$ block consists of t independent constant coefficients.

At an actual matrix, inclusion–exclusion identifies h_S with the sum of the matchings using exactly the row set S . For each i , direct counting gives

$$\sum_{a=1}^s \partial_{ia} M_{t,s,d} = (s-d+1)c_{\{i\}}. \quad (16)$$

A matching of size $d-1$ avoiding row i is counted once for each unused column. Since $s-d+1 \neq 0$ in the ground field, every $c_{\{i\}}$ vanishes on the critical locus. Imposing this zero block on the formal coefficient image leaves dimension $R_{\text{crit}}(t, d) = R(t, d) - t$.

Choose coordinates on each remaining disjointness image and use them to reconstruct the coefficients of (15). Together with the $s(d-2)$ column elementary invariants used above, these coordinates generate a polynomial parameter ring of dimension $R_{\text{crit}}(t, d) + s(d-2)$. Its universal column incidence still contains every critical matrix with its actual coefficient data. The leading part of Q_i is unchanged, so the same degree-reducing monic relations make its coordinate ring finite over this parameter ring. Surjecting onto the critical-locus coordinate ring proves (10).

For use in the parameter estimates, the count has the explicit forms

$$\begin{aligned} R_{\text{crit}}(t, 2h) &= 2 \sum_{j=1}^{h-1} \binom{t}{j} + \binom{t}{h} - t, \\ R_{\text{crit}}(t, 2h+1) &= 2 \sum_{j=1}^h \binom{t}{j} - t. \end{aligned}$$

The dimension calculation is exact for the formal coefficient image. It does not assert that actual critical matrices fill that image, or that the resulting upper bound on their dimension is sharp.

A.3 The selector identity

Use the square-zero algebra $\mathbb{C}[z_1, \dots, z_r]/(z_1^2, \dots, z_r^2)$ and set $y_h = \sum_{i \in R_h} z_i$. Then $y_h^{t+1} = 0$. The product of the linear forms associated with the constant columns of U is

$$\mathcal{S} := \prod_{h=1}^b y_h^{t-d} \prod_{h=2}^b ((y_1 + \dots + y_{h-1})^d + \theta y_h^d), \quad \theta = (-1)^{d+1} 2^d. \quad (17)$$

This follows from $\prod_{j=0}^{d-1} (S + 2\zeta^j Y) = S^d + \theta Y^d$.

In this truncated algebra, induction on the number of blocks gives

$$\mathcal{S} = \sum_{h=1}^b \lambda_h y_h^{t-d} \prod_{j \neq h} y_j^t. \quad (18)$$

For the induction, every term has exactly one block of exponent $t-d$, with all other old blocks of exponent t . In the next factor, $(y_1 + \dots + y_{h-1})^d$ can survive only by taking all d powers from that one unsaturated block. It saturates the old block and leaves the new one unsaturated. The other term, θy_h^d , keeps the old unsaturated block and multiplies its coefficient by θ . Thus old coefficients are multiplied by θ , and the new coefficient is their previous sum. Starting from 1, their sums are $(1+\theta)^{h-1}$, yielding the displayed formulas for λ_h in the main text.

For a set I of d omitted rows, the coefficient of $\prod_{i \notin I} z_i$ in the product of the column forms is $\text{per}(U_{[r] \setminus I, \bullet})$. Equation (18) makes this zero unless I lies within one block R_h . In that case the coefficient is $\lambda_h(t-d)!(t!)^{b-1}$, independent of the particular I . Finally expand the full permanent

along its $s - d$ all-ones bottom rows. Exactly d columns from the X block remain for the top rows, and the bottom contribution is $(s - d)!$. Summing the remaining size- d matchings gives

$$\text{per} \begin{pmatrix} X & U \\ 1 & 0 \end{pmatrix} = (s - d)!(t - d)!(t!)^{b-1} \sum_{h=1}^b \lambda_h M_{t,s,d}(X^{(h)}),$$

as required. Both θ and $1 + \theta$ are nonzero, so every weight used in the critical-locus product is nonzero.

B Algebraic transfer with polar multiplicities

This appendix gives a second proof of the smooth-target transfer and records its characteristic restrictions. The determinantal incidence and its multidegree are still the credited mechanism of [3]. The additional argument below replaces persistence of reduced complex points by conservation of isolated complete-intersection length. In particular, it does not assume that the Gauss map is separable.

Lemma B.1 (Local conservation). *Let R be a complete discrete valuation ring with algebraically closed residue field K and uniformizer t . Let $F_1, \dots, F_k \in R[x_1, \dots, x_k]$. Suppose their special fiber has a finite set of isolated points z , with local lengths μ_z . There are finite flat neighborhoods of these points in the common zero scheme, of combined rank $\sum_z \mu_z$ over R . Any function nonzero at a selected special point is a unit on its corresponding finite local factor.*

Proof. Take the quasi-finite open locus around the selected points. Zariski's Main Theorem embeds it into a finite R -scheme. Since R is henselian, this finite scheme splits into factors indexed by its closed-fiber points. A factor whose closed point lies in the chosen open lies entirely in that open: every nonempty closed subset of a finite local factor contains its closed point. These factors give finite neighborhoods in the original zero scheme.

At each selected point the ambient local ring is regular of dimension $k + 1$. The k equations followed by t form a system of parameters, because their quotient has finite length. A system of parameters in a regular local ring is a regular sequence. Thus t is a non-zero-divisor after imposing the F_i . Each finite local factor is torsion-free, hence free, over R , with rank equal to its special-fiber length. The unit assertion follows by reduction to the closed point. These are standard complete-intersection and henselian arguments; compare [8, Sections 10.104, 10.153 and Lemma 93.17.4]. $\square$

Proposition B.2. *Let K be algebraically closed, in any characteristic. If f is a homogeneous degree- d form in $k \geq 3$ variables and $V(f) \subset \mathbb{P}_K^{k-1}$ is smooth, then*

$$\overline{\text{dc}}_K(f) \leq m \implies d(d-1)^{k-2} \leq B(m, k).$$

Consequently $\overline{\text{dc}}_K(f) \geq (k-1)(d-1)/(4e)$.

Proof. Set $X = V(f)$. Smoothness implies that the partial derivatives have no common zero on X . They define a morphism

$$\gamma : X \longrightarrow \mathbb{P}^{k-1}, \quad \gamma^* \mathcal{O}(1) = \mathcal{O}_X(d-1).$$

This morphism is finite onto its image: a positive-dimensional fiber would contain a projective curve on which the ample bundle $\mathcal{O}_X(d-1)$ had degree zero. Hence $k-2$ general hyperplanes in the target pull back to a finite scheme of length

$$\delta = \int_X c_1(\mathcal{O}_X(d-1))^{k-2} = d(d-1)^{k-2}.$$

It is cut out by directional derivatives $D_1f, \dots, D_{k-2}f$. Choose ℓ nonzero on its finite support. In the affine normalization $\ell = 1$, the equations are

$$f = 0, \quad \ell - 1 = 0, \quad D_1f = \dots = D_{k-2}f = 0.$$

This is a zero-dimensional complete intersection of length δ . It can be nonreduced. At every point, however, some first partial of f is nonzero. We have used the gradient only on X ; no division by d in Euler's identity is needed.

The affine size- m determinants form a constructible image in coefficient space. Algebraic curve selection at a point of its closure, followed by a finite function-field extension, normalization, and completion, gives a complete discrete valuation ring R with residue field K and a polynomial

$$g \in R[x_1, \dots, x_k], \quad g \bmod t = f, \quad g = \det A \text{ over } \text{Frac}(R).$$

The matrix entries may have poles; only the coefficients of g are required to be regular. A constant family suffices when f already has an exact representation.

Apply Lemma B.1 to $g, \ell - 1, D_1g, \dots, D_{k-2}g$. It retains a generic-fiber scheme of length δ . A first partial nonzero at each selected special point stays a unit on its finite factor. Thus every retained generic matrix value has corank one. Extend the generic field to an algebraic closure, which does not change length, and choose Λ injective on all of the finitely many matrix images.

The Schur-complement elimination from Section 3 is scheme-theoretic: it divides only by units and identifies the kernel incidence locally with $g = 0$. The identity $\text{adj}(A) = \det(B)vu^T$ identifies the lifted polar equations with $D_jg/\det(B)$. Hence it preserves the complete local polar schemes, including their nilpotents and total length.

Finally perturb the multihomogeneous incidence equations by a new formal parameter to general sections of the same globally generated line bundles. Their proper intersection has length

$$[H^k U^{m-1} V^{m-1}] H(H + U)^m (H + V)^{m-1} (U + V)^{k-2} = B(m, k).$$

Apply Lemma B.1 locally on affine charts of the smooth product of projective spaces. Every original isolated component retains its length. Thus $\delta \leq B(m, k)$ even when the original incidence has excess components elsewhere. The numerical estimate in Section 3 is characteristic-independent. $\square$

Corollary B.3 (Large characteristic). *Let K be algebraically closed of characteristic $p > n$. Both inequalities of Theorem 1.1 hold over K in their respective ranges $n \geq 64$ and $n \geq 512$.*

Proof. The matching finite-module proof divides by $(d - 1)!$ and $t - d + 1$. The compression additionally uses $s - d + 1$ and the ranks of inclusion matrices on a t -element set. All these arguments apply for $p > \max(t, s)$, by Appendix A.2. The selector uses roots of unity of order $d < p$ and the nonzero factor $(s - d)!(t - d)!(t!)^{b-1}$. For both parameter choices in the proof of Theorem 1.1, $t, s \leq n$ and $2^d < n$; hence θ and $1 + \theta$ are nonzero modulo p . The smooth-slice incidence argument applies because $p > d$, so Euler's identity identifies the critical cone with the hypersurface singular cone. Proposition B.2 and the same integer estimates prove the claim. $\square$

This is not a superquadratic result in any fixed positive characteristic. In particular, removing the analytic transfer does not remove the selector's factorial obstruction.

Multiplicity controls. For $d = p + 1$, the smooth Fermat form in k variables has inseparable polar equations $x_i^p = 0$ on a suitable affine chart. The polar length is $(p + 1)p^{k-2}$, whereas its support has only $p + 1$ points. The triangular deformation $x_i^p + t = 0$ retains this length. A smooth characteristic-two conic also checks the case $p \mid d$, where the ambient gradient can have a base point outside the hypersurface. These exact finite controls test why length, rather than distinct-point count, is needed; the proof above supplies the general assertion.

C A charge for non-skew multiplication gates

This appendix extends the polar count to a circuit model with a stated budget on non-skew multiplication gates. A binary multiplication gate is called *non-skew* if neither child is an input variable or field constant. Other multiplication gates are skew. Circuits are commutative and division-free; arbitrary fan-out and cancellation are allowed. The related noncommutative restrictions studied in [9] are different models, and their lower bounds are not used.

Theorem C.1 (Circuit-state polar count). *Let f be a homogeneous degree- d form in $k \geq 3$ variables with smooth projective hypersurface. Suppose f is a fixed affine restriction of a coefficient limit of outputs of circuits with S internal binary gates, N input variables, and at most q non-skew multiplication gates, where $0 \leq q \leq S$. A larger stated budget may always be replaced by S . Put $M = S + N + 1$. Then*

$$d(d-1)^{k-2} \leq T_q(M, k),$$

where $T_0(M, k) = B(M, k)$ and, for $q \geq 1$,

$$T_q(M, k) \leq 2^{q+k-2} \binom{q+k-3}{k-2} \binom{2M-1}{k-1}. \quad (19)$$

In particular, for every fixed $C \geq 0$, the budget $q \leq C(k-1)$ implies $M = \Omega_C(kd)$ as d tends to infinity.

Corollary C.2. *For each fixed constant C , a division-free circuit computing per_n with at most Cn^2 non-skew multiplication gates has $\Omega_C(n^2 \log n)$ internal gates. The same holds for coefficient approximation with these size and non-skew-gate budgets.*

Proof. Use the fixed smooth restriction from the proof of the first part of Theorem 1.1, taking $t = \lfloor \log_2 n \rfloor$, $d = \lfloor t/2 \rfloor$, $k = \lfloor n^2/16 \rfloor$ for all sufficiently large n , and $N = n^2$ original input registers. Affine restriction changes their labels to affine forms but preserves their role as skew operands in the construction below. Theorem C.1 gives $S + N + 1 = \Omega_C(n^2 \log n)$; subtracting $N + 1$ proves the assertion for all sufficiently large n . $\square$

State equations and their adjoints

Introduce one state w for each original input variable and each internal gate, for a total of $M - 1$ states. After restriction, the input equations are $w_i = L_i(x)$ with L_i affine. In topological order, each addition, subtraction, or skew multiplication has an equation affine in the states, with coefficients affine in x . For a skew product, replace its input child by the affine label $L_i(x)$; retain the state register for the other operand. Constants require no state register. Each non-skew product has an equation

$$w_g - w_a w_b = 0.$$

The Jacobian of these $M - 1$ graph equations with respect to the states is triangular with diagonal one. Thus they define the graph of the computation, including every use of every shared value.

Add the output equation $w_{\text{out}} = 0$. On $\mathbb{P}_{[t:x]}^k \times \mathbb{P}_{[v_0:w]}^{M-1}$, the $M - q$ linear-state equations, including the output, become forms F_i of bidegree $(1, 1)$. Examples are

$$tv_i - \hat{L}_i(x)v_0, \quad tv_g - \hat{L}_i(x)v_a, \quad tv_{\text{out}}.$$

For the other q equations use

$$tQ_j(v) = t(v_g v_0 - v_a v_b),$$

of bidegree $(1, 2)$. The added factors t have no effect on the chart $t = v_0 = 1$. Their extra components at infinity are permitted.

At a zero of the output with nonzero differential, the Jacobian of all M constraints in the $M - 1$ state directions has rank $M - 1$. Its left kernel is a unique multiplier line. Write the multipliers as u_i for the F_i and λ_j for the tQ_j . The multiplier of the output equation is nonzero: otherwise the invertible graph Jacobian forces every multiplier to vanish. Normalize that multiplier to one. Implicit differentiation of the graph equations shows that

$$S_y = \sum_i u_i (F_i)_y + \sum_j \lambda_j (tQ_j)_y$$

vanishes for all state directions, and its x -components are, up to sign, the differential of the output. The state adjoints sum all contributions from fan-out; no parse-tree expansion occurs.

Choose a simple polar fiber as in Proposition 3.1. It has $\delta = d(d - 1)^{k-2}$ points. At each point the graph equations eliminate the states and the adjoint equations eliminate the normalized multipliers, by invertible Jacobians. The resulting local system is the original square polar system. Every lifted point is therefore isolated and reduced. The same is true for the δ points persisting in any sufficiently close polynomial circuit output.

A bundle count that respects reuse

Write H, V for the hyperplane classes of $Y = \mathbb{P}^k \times \mathbb{P}^{M-1}$ and use the quotient projective bundle

$$\pi : \mathbb{P}(E') \longrightarrow Y, \quad E' = \mathcal{O}_Y^{\oplus(M-q)} \oplus \mathcal{O}_Y(V)^{\oplus q}, \quad L = c_1(\mathcal{O}_{\mathbb{P}(E')}(1)).$$

The coordinates u_i have class L , whereas the λ_j have class $L - V$. Hence a state-adjoint equation has class $L + H$, and an x -directional adjoint equation has class $L + V$. The latter has no quadratic-constraint contribution, since Q_j is independent of x ; it is nevertheless a section of the stated line bundle. The homogenizing coordinate t is not one of these x directions.

The incidence has $M - q$ equations of class $H + V$, q of class $H + 2V$, $M - 1$ state adjoints of class $L + H$, $k - 2$ polar adjoints of class $L + V$, and one equation $\ell(x) - t$ of class H . Their number is $k + 2M - 2 = \dim \mathbb{P}(E')$. All these bundles are globally generated: E' is globally generated, as are its twists by H and V . General perturbations of the sections thus have proper zero-dimensional intersection. The isolated points just constructed persist, giving the upper bound

$$T_q(M, k) = \int_{\mathbb{P}(E')} H(H + V)^{M-q} (H + 2V)^q (L + H)^{M-1} (L + V)^{k-2}. \quad (20)$$

This bound concerns the original system's isolated points and allows excess components elsewhere. In particular, it does not assume properness of the incidence at infinity.

For $q = 0$ the bundle is trivial and (20) is $B(M, k)$ after exchanging the two kernel-factor names. For $q \geq 1$, the projective-bundle formula gives

$$\pi_*(L^{M-1+a}) = \binom{q+a-1}{a} V^a \quad (a \geq 0), \quad \pi_*(L^j) = 0 \quad (j < M-1).$$

For precision about conventions, these formulas follow from $L^{M-q}(L - V)^q = 0$ and $\pi_*(L^{M-1}) = 1$ by induction; see [8, Sections 27.21 and 42.36]. No sign convention for Segre classes is implicit.

All coefficients in the integrand and pushforward are nonnegative. Bound $(H + 2V)^q$ coefficient-wise by $2^q(H + V)^q$. If i copies of H are chosen from $(H + V)^M$ and c from $(L + H)^{M-1}$, the

required H degree is equivalent to $i + c = k - 1$. Choosing j copies of L from $(L + V)^{k-2}$ leaves a pushforward contribution bounded by

$$\sum_{j=c}^{k-2} \binom{k-2}{j} \binom{q+j-c-1}{j-c} \leq 2^{k-2} \binom{q+k-3}{k-2}.$$

The remaining sum is

$$\sum_{i+c=k-1} \binom{M}{i} \binom{M-1}{c} = \binom{2M-1}{k-1},$$

which proves (19). The count is nondecreasing in q : both the constraint coefficients and the pushforward coefficients increase. An upper budget on q therefore suffices.

Fixed-size circuit families form a finite union of constructible polynomial images. The simple target polar points persist under coefficient approximation, with no boundedness assumption on circuit parameters or state values. This proves the border assertion and completes the proof of Theorem C.1.

An explicit tradeoff and its limit

Taking $(k-1)$ -st roots and using the standard binomial estimates gives the convenient bound, for all $q \geq 0$,

$$M \geq \frac{1}{2} + \frac{(k-1)(d-1)}{4e^2 2^{q/(k-1)}(1+q/(k-2))}. \quad (21)$$

For $q = 0$, Proposition 3.1 gives a stronger constant. For $q \geq 1$, use

$$\binom{q+k-3}{k-2}^{1/(k-1)} \leq e \left(1 + \frac{q}{k-2}\right), \quad \binom{2M-1}{k-1}^{1/(k-1)} \leq \frac{e(2M-1)}{k-1}.$$

If $q \leq C(k-1)$, the denominator of (21) is bounded by a constant depending only on C , proving the last assertion of the theorem.

This charge tolerates $O(k)$ non-skew gates but weakens as their number grows. For example, repeated squaring computes $\sum_{i=1}^k x_i^{2^a}$ using $k(a-1)$ non-skew squarings after the first squares. Here q/k grows with $\log d$, outside the fixed- C regime. Thus the argument respects the earlier counterexample to a linear charge of smooth-slice score to unrestricted circuit size. Corollary C.2 is a restricted circuit result, not an unrestricted $\Omega(n^2 \log n)$ circuit bound.

The accompanying finite checker independently expands 416 intersection coefficients, checks 364 monic projective-bundle reductions and 32 repeated-squaring controls, and verifies 35 full local incidence Jacobians, including 77 multiply-used internal states. These checks support the displayed identities and local elimination; the preceding argument proves the uniform statement.

D Generic output levels and valid divisions

This appendix extends the state construction to valid divisions and formal rational-output approximation over $\mathbb{C}$. It is separate from the main determinantal theorem and retains an explicit gate budget. In particular, it does not extend the two-kernel determinant incidence to nonzero determinant levels.

Theorem D.1. *Let a circuit over $\mathbb{C}((\varepsilon))$ have S internal binary gates, N inputs, arbitrary fan-out, and valid divisions. Replace divisions by nonzero field constants by scalar multiplications. Count in q every remaining division and every non-skew multiplication, as defined in Appendix C; thus $0 \leq q \leq S$. Suppose its rational output satisfies*

$$r_\varepsilon(X) = F(X) + O(\varepsilon) \quad \text{in } \mathbb{C}(X)((\varepsilon)),$$

where F is a polynomial. Suppose a fixed affine restriction $F(Ax)$ is a homogeneous degree- d form f in $k \geq 3$ variables, with $d \geq 2$ and smooth projective hypersurface. Set $M = S + N + 1$. Then

$$d(d-1)^{k-1} \leq T_q(M, k+1). \quad (22)$$

Consequently,

$$M \geq \frac{1}{2} + \frac{k(d-1)}{4e^2 2^{q/k}(1+q/(k-1))}. \quad (23)$$

For every fixed C , the budget $q \leq Cn^2$ therefore forces $S = \Omega_C(n^2 \log n)$ for such an approximation to per_n .

Here A denotes an affine map, including its constant term. The output need not be polynomial for nonzero parameter. The gate graph and its input roles are fixed; constants may be Laurent series.

A full-dimensional polar target

Smoothness and Euler's identity imply $\text{Crit}(f) = \{0\}$. The gradient defines a finite morphism

$$\gamma : \mathbb{P}^{k-1} \longrightarrow \mathbb{P}^{k-1}, \quad \gamma^* \mathcal{O}(1) = \mathcal{O}(d-1),$$

of degree $(d-1)^{k-1}$. In characteristic zero it is generically étale. A general direction a avoids both its branch locus and $\gamma(V(f))$. Its projective fiber has $(d-1)^{k-1}$ distinct points at which f is nonzero. For each nonzero c , every such point has exactly d scalar representatives with $f = c$. Thus the map

$$\Psi_f : x \longmapsto (f(x), [\nabla f(x)])$$

has a general fiber of $\delta = d(d-1)^{k-1}$ simple points. In a target chart $a_k = 1$ the corresponding square system is

$$f - c = 0, \quad f_{x_i} - a_i f_{x_k} = 0 \quad (1 \leq i < k). \quad (24)$$

Its Jacobian is invertible: the direction equations are étale in projective coordinates, while the remaining radial derivative is dc , which is nonzero.

For any nonzero polynomial E_0 , the closure of the image of $\{E_0 = 0\}$ under this map, on its domain, has dimension at most $k-1$. The target has dimension k . A general (c, a) consequently has all its δ preimages outside $\{E_0 = 0\}$. Allowing the output level to vary is essential: a circuit with output $f^2/f = f$ is invalid at every point of the zero level.

A ramified translation of the fixed slice

We give the argument because direct substitution of a fixed slice can annihilate internal divisors. Represent each intermediate value as a quotient of polynomials in X over $\mathbb{C}((\varepsilon))$. There is a finite list of nonzero polynomials whose nonvanishing ensures every division is valid; include all these numerators and denominators, and an output denominator. Normalize each polynomial H by a

Laurent scalar so that $H \in \mathbb{C}[[\varepsilon]][X]$ and its reduction H_0 is nonzero. The degrees are finite. Choose a constant vector b such that no $H_0(A(0) + \tau b)$ is identically zero, and choose an integer R larger than all these degrees. These choices exist because there are finitely many proper exceptional algebraic sets of vectors b .

Substitute $\varepsilon = \tau^R$ and $X = Ax + \tau b$. The τ -adic valuation of each $H_0(Ax + \tau b)$ is at most $\deg H_0 < R$. Terms of positive ε order have τ order at least R , so cannot cancel this leading term. Every original division therefore remains valid as a rational function after substitution. Neither S nor q changes: an original input operand is still represented by its affine label in the state construction.

Write the output as N/D , normalizing the denominator to have coefficientwise valuation zero. The approximation hypothesis and the Gauss valuation imply $N - FD \in \varepsilon\mathbb{C}[[\varepsilon]][X]$. If $a < R$ is the valuation of the substituted denominator, write it as $\tau^a E$, with $E \in \mathbb{C}[[\tau]][x]$ and $E_0 \neq 0$. The new rational output has the form

$$g(\tau, x) = F(Ax + \tau b) + \tau^{R-a} \frac{H(\tau, x)}{E(\tau, x)}, \quad H \in \mathbb{C}[[\tau]][x]. \quad (25)$$

On $E_0 \neq 0$, the output and its x -derivatives reduce to f and its derivatives.

Choose (c_0, a_0) so that the δ simple points of (24) avoid $E_0 = 0$. For every $b' \in \mathbb{C}^k$, the target $(c_0, a_0) + \tau b'$ has δ distinct simple Hensel lifts for the system with f replaced by g . At these lifts E and g_{x_k} are units. Internal canceled poles still require a separate check. Over $K = \mathbb{C}((\tau))$, let U be the nonempty open set where all original divisions are valid and let $V = \{E \neq 0, g_{x_k} \neq 0\}$. On V the map

$$\Psi_g = (g, g_{x_1}/g_{x_k}, \dots, g_{x_{k-1}}/g_{x_k})$$

is a morphism to $\mathbb{A}_K^k$. The closure of the image of $V \setminus U$ has dimension at most $k - 1$. The set $(c_0, a_0) + \tau\mathbb{C}^k$ is Zariski dense over K : an invertible affine change reduces this to the density of the infinite Cartesian set $\mathbb{C}^k$ for polynomials over K . Choose b' avoiding that bad image. All the retained Hensel lifts then belong to U .

Shared states, division pivots, and the count

Introduce the $M - 1$ input and gate states as in Appendix C. A division $w_g = w_a/w_b$ has equation $w_g w_b - w_a = 0$. At every retained valid point the graph Jacobian in topological state order is triangular, with pivot one at ordinary gates and nonzero pivot w_b at a division. It is invertible. Add $w_{\text{out}} - c(\tau) = 0$. The state adjoints have a unique multiplier line whose output multiplier is nonzero. Normalizing it to one and eliminating states and adjoints recovers (24) with $g, c(\tau), a(\tau)$ in place of f, c, a . Implicit differentiation includes every outgoing use of a shared gate. The lifted incidence points are isolated and reduced.

On $Y = \mathbb{P}_H^k \times \mathbb{P}_V^{M-1}$, homogenize the $M - q$ linear-state constraints to class $H + V$. The q quadratic-state constraints have class $H + 2V$; for division use $t(v_g v_b - v_a v_0)$. The added factor t is a unit on the chosen affine chart. Use the same quotient bundle $E' = \mathcal{O}^{M-q} \oplus \mathcal{O}(V)^q$ and tautological class L as in Appendix C. There are $M - 1$ state adjoints of class $L + H$ and now $k - 1$ polar adjoints of class $L + V$. There is no equation $\ell = 1$. The total number of equations is $k + 2M - 2 = \dim \mathbb{P}(E')$. Their bundles are globally generated, so the same isolated-point conservation argument gives

$$\delta \leq \int_{\mathbb{P}(E')} (H + V)^{M-q} (H + 2V)^q (L + H)^{M-1} (L + V)^{k-1} = T_q(M, k + 1).$$

For the last identity, the additional H in (20) cancels the additional required base H degree when k is replaced by $k + 1$. All counts may be made over an algebraic closure of K ; no boundedness of

gate constants or state values is needed. Substituting $k + 1$ into (19) and the root estimate proves (23). Applying the fixed permanent slice and subtracting $N + 1$ proves the final assertion of the theorem.

The independent finite controls check 294 shifted coefficient identities, 24 full rational-circuit incidence Jacobians, two canceled-pole examples, and eight Fermat level counts. They reject 24 deliberately incorrect unit-pivot substitutions. These are implementation checks, not a substitute for the argument above. No extension of this generic-level proof to positive characteristic is asserted here.

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